

Core Algebra Practice

1. Simplify: $-5x + 5 + 6x - 4$.
2. Expand and simplify: $(x + 3)(x + 7)$.
3. Factor completely: $x^2 - 16$.
4. Complete the square: $x^2 + 10x + 25$.
5. Evaluate $-6x^2 - 3x + 6$ when $x = -1$.
6. Solve the inequality: $4x + 8 < -12$.
7. Solve: $|5x + 40| = 40$.
8. Simplify: $\frac{x^4 x^{10}}{x^{10}}$.
9. Solve and state the restriction: $\frac{12}{x+6} = 2$.
10. Solve and check: $\sqrt{x-3} = 5$.
11. Make a the subject of $v = u + at$.

Solutions

1. Combine like terms: $x + 1$.
2. After distributing: $x^2 + 10x + 21$.
3. The zeros are $-4, 4$, so $x^2 - 16 = (x + 4)(x - 4)$.
4. $x^2 + 10x + 25 = (x + 5)^2$.
5. Substitution gives 3.
6. $x < -5$.
7. Use both cases to obtain $x = -16, 0$.
8. Add numerator exponents and subtract the denominator exponent: x^4 , where $x \neq 0$.
9. Restriction: $x \neq -6$. Solving gives $x = 0$.
10. Square both sides: $x - 3 = 25$, so $x = 28$.
11. $a = \frac{v-u}{t}$.